energy gamma rays, and not into magnetic energy and electron kinetic energy. Thus, matter/antimatter annihilation appeared to be a very unsatisfactory way to energize a quasar.

One other possibility remained: *gravity.* The implosion of a normal star to form a neutron star or a black hole might, conceivably, convert 10 percent of the star's mass into magnetic and kinetic energy—though precisely how was unclear. If it managed to do so, then the implosions of $10 \times 10^7 = 10^8$ (100 million) normal stars might provide a quasar's energy, as would the implosion of a single, hypothetical, *supermassive star* 100 million times heavier than the Sun. [The correct idea, that the gigantic black hole produced by the implosion of such a supermassive star might itself be the engine that powers the quasar, did not occur to anybody in 1963. Black holes were but poorly understood. Wheeler had not yet coined the phrase "black hole" (Chapter 6). Salpeter and Zel'dovich had not yet realized that gas falling toward a black hole could heat and radiate with high efficiency (Chapter 8). Penrose had not yet discovered that a black hole can store up to 29 percent of its mass as rotational energy, and release it (Chapter 7). The golden age of black-hole research had not yet begun.]

The idea that the implosion of a star to form a black hole might energize quasars was a radical departure from tradition. This was the first time in history that astronomers and astrophysicists had felt a need to appeal to effects of general relativity to explain an object that was being observed. Previously, relativists had lived in one world and astronomers and astrophysicists in another, hardly communicating. Their insularity was about to end.

To foster dialogue between the relativists and the astronomers and astrophysicists, and to catalyze progress in the study of quasars, a conference of three hundred scientists was held on 16–18 December 1963, in Dallas, Texas. In an after-dinner speech at this First Texas Symposium on Relativistic Astrophysics, Thomas Gold of Cornell University described the situation, only partially with tongue in cheek: "[The mystery of the quasars] allows one to suggest that the relativists with their sophisticated work are not only magnificent cultural ornaments but might actually be useful to science! Everyone is pleased: the relativists who feel they are being appreciated and are experts in a field they hardly knew existed, the astrophysicists for having enlarged their domain, their empire, by the annexation of another subject—general relativity. It is all very pleasing, so